

Trainings ▾

General Information

All maintenance procedures listed for previous intervals **must** also be performed.

For your convenience, listed below are the section numbers that contain specific instructions for performing the maintenance.

Maintenance Procedures at Daily Interval (Section 3)

- Air Intake Piping - Maintenance Check
- Air Cleaner Precleaner - Maintenance Check
- Air Cleaner Restriction - Maintenance Check
- Air Tanks and Reservoirs - Maintenance Check
- Engine Coolant Level - Maintenance Check
- Fuel-Water Separator - Drain
- Engine Lubricating Oil Level - Maintenance Check
- Centinel™ Oil Level - Maintenance Check

Maintenance Procedures at 250 Hours or 6 Months (Section 4)

- Fuel Lift Pump Inlet Screen - Clean
- Fuel Filter (Spin-on Type) - Replace and Prime
- Lubricating Oil and Filters - Change¹
- Lubricating Oil Analysis - Maintenance Check (Centinel™ Equipped Engines **Only**)
- Crankcase Breather Tube - Maintenance Check
- Supplemental Coolant Additive - Maintenance
- Coolant Filter - Replace
- Drive Belts - Maintenance Check
- Fan, Cooling - Maintenance Check
- Charge Air Cooler (CAC) - Maintenance Check
- Air Cleaner Element – Change

Maintenance Procedures at 2000 Hours or 1 Year (Section 5)

- Engine Steam Cleaning - Clean
- Radiator Hoses - Maintenance Check
- Air Compressor Discharge Lines - Maintenance Check
- Engine Mounts - Maintenance Check
- Crankshaft - Maintenance Check

- Overhead Set - Adjust
- Engine Oil Heater - Maintenance Check
- Coolant Heater - Maintenance Check
- Fan Drive Idler Pivot Arm Assembly - Maintenance Check
- Drive Belt, Cooling Fan - Adjust

Maintenance Procedures at 4000 Hours or 2 Years (Section 6)

- Fuel Lift Pump - Replace
- Fuel Tank - Drain (Refer to OEM service manual for draining fuel tank and cleaning procedures)

Maintenance Procedures at 6000 Hours or 2 Years (Section 7)

- Cooling System - Clean
- Fan Hub Belt Driven - Maintenance Check
- Water Pump - Maintenance Check
- Turbocharger - Maintenance Check
- Vibration Damper, Viscous - Maintenance Check
- Air Compressor Unloader Valve Assembly - Maintenance Check

Maintenance Procedures at 10000 Hours or 2 Years (Section 8)

- Turbocharger - Replace²

Notes:

1. The lubricating oil and filter change intervals can be extended, based on the engine's horsepower rating, oil pan capacity, and fuel consumption rate. Reference the Oil Drain Intervals information in this section for specific details. For Centinel equipped engines, the oil filter must be replaced at 1,000 hours.
2. For engine operating with a high altitude calibration, the turbocharger **must** be replaced at 10,000 hours or engine half life, whichever comes first.

Oil Drain Intervals

⚠ CAUTION ⚠

Improperly applied extended oil and filter change intervals will decrease engine life due to factors such as corrosion, deposits, and wear.

Centinel™

With a continuously operating Centinel™ System, the oil drain interval can be extended until the oil analysis requires the oil to be changed or the oil is known to be contaminated. Oil analysis is required at 250 hour intervals when using the Centinel™ System and **must** include soot measurement. Fleetguard® oil analysis kit, Part Number CC 2543, meets this requirement. The oil degradation limits can be found in the Centinel™ oil analysis recommendations table in Section V of this manual.

Premium oil filters, such as Fleetguard® LF 9000 series filters, or equivalent, are also required when using Centinel™ with spin-on filters.

There are three recommended methods used to determine the proper oil and filter change interval:

- Fixed Hours Method (based on fixed hours, or months; whichever occurs first)
- Chart Method (based on known fuel consumption rates)
- Centinel™ system (based on lubricating oil analysis).

Reference the table below for the Fixed Hours Method to determine the lubrication oil and filter change interval, unless using the chart method.

Fixed Hours Method Oil Intervals				
Application	Oil Filters	Sump Capacity	Oil Grade	Change Interval
Industrial	LF670 full-flow and LF777 bypass	76 liters [20 gal]	Standard or Premium	250 hours or 6 months
Industrial	LF670 full-flow and LF777 bypass	133 liters [35 gal]	Standard or Premium	250 hours or 6 months
Industrial	LF14000NN full-flow	76 liters [20 gal]	Standard or Premium	250 hours or 12 months
Industrial	LF14000NN full-flow	133 liters [35 gal]	Standard	250 hours or 12 months
Industrial	LF14000NN full-flow	133 liters [35 gal]	Premium	500 hours or 12 months
Power Generation	LF670 full-flow and LF777 bypass	76 liters [20 gal]	Standard or Premium	250 hours or 12 months
Power Generation	LF670 full-flow and LF777 bypass	133 liters [35 gal]	Standard or Premium	250 hours or 12 months
Power Generation	LF14000NN full-flow	76 liters [20 gal]	Standard or Premium	250 hours or 12 months

Fixed Hours Method Oil Intervals

Application	Oil Filters	Sump Capacity	Oil Grade	Change Interval
Power Generation	LF14000NN full-flow	133 liters [35 gal]	Standard	250 hours or 12 months
Power Generation	LF14000NN full-flow	133 liters [35 gal]	Premium	500 hours or 12 months

Refer to Procedure 018-003 (</qs3/pubsys2/xml/en/procedures/57/57-018-003.html>) in Section V for determining the oil performance classification.

⚠ CAUTION ⚠

Inaccurate fuel consumption records can result in incorrect oil change intervals that can result in premature engine wear and/or deposits.

⚠ CAUTION ⚠

Do not extend oil change intervals beyond the maximum point shown when fuel consumption rates are less than the minimum shown. Doing so can result in shortened engine life due to oil degradation processes associated with lightly loaded engines.

Chart Method:

There has been an increased emphasis on extended oil drain intervals for high-horsepower engines. This is one way that longer intervals can be achieved. The chart method, when coupled with a thorough oil analysis program, is a viable option accepted by Cummins Inc.

Do **not** extend oil and filter change intervals beyond the Fixed Hour Method, unless the chart method is used.

The chart method is recommended to provide the lowest total cost of operation while still protecting the engine.

The information listed below is required when using the chart method to determine the correct oil and filter change interval for the engine.

- Oil sump capacity
- Average fuel consumption rate
- Classification of oil as standard or premium.

Determine the engine oil sump capacity:

Sump capacity in gallons can be determined by knowing the volume of the oil required to touch the high-level mark on the dipstick. Use the following procedure for oil sump capacities. Refer to Procedure 018-017 in Section V (</qs3/pubsys2/xml/en/procedures/57/57-018-017.html>). It is imperative that the sump volume be known when using the chart method.

Determine the fuel consumption rate:

To use the chart method effectively, accurate fuel consumption records **must** be kept and maintained. Fuel consumption information **must** be in liters per hour (lph) or gallons per hour (gph) units to use the charts. Fuel consumption rates can change over time due to an increasing or decreasing engine duty cycle. Accurate records are essential in determining average fuel consumption during a given oil change interval.

To read the chart:

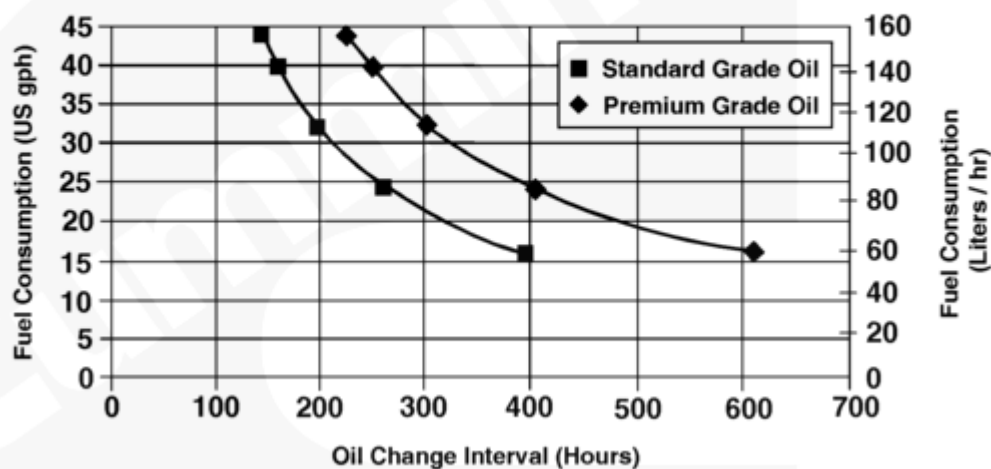
Select the chart representing the correct oil sump capacity. The left vertical axis of the chart represents fuel consumption in gallons per hour (gph) and the right vertical axis represents fuel consumption in liters per hour (lph). Determine the intersection point of the fuel consumption rate on the applicable interval curve by drawing a horizontal line. From this point, draw a vertical line down until it intersects with the horizontal axis representing hours. This point represents the acceptable oil change interval.

Oil Filters:

Premium filters are recommended, in conjunction with premium grade oils, when using the premium grade oil curves to determine oil change intervals. Premium filters contain synthetic media materials to provide more efficient filtration for the entire service life and to extend the media life over conventional cellulose media. Premium filters are made with synthetic media and have the StratPore™ designation on the outside of the filter. StrataPore™ filters have the efficiency, capacity, and strength needed for this extended service.

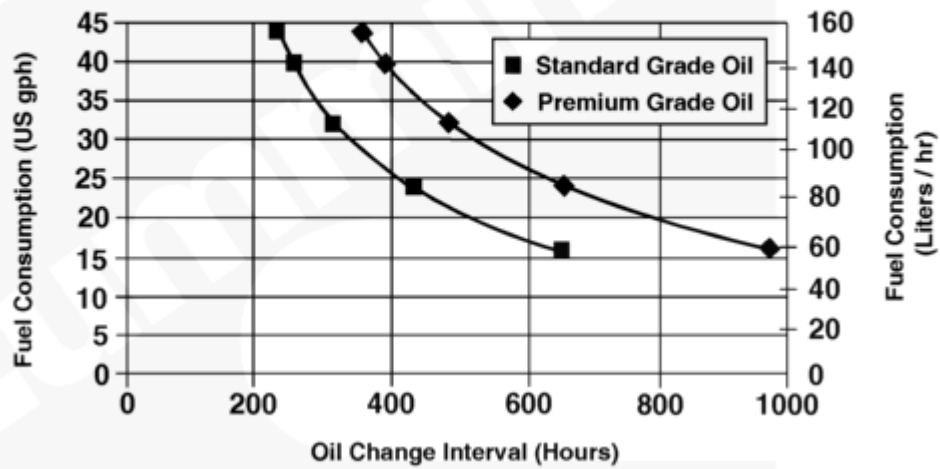
©Cummins Inc

QST30 Oil Change Interval 76 Liter (20 US Gallon) Sump



07a00130

QST30 Oil Change Interval 132 Liter (35 US Gallon) Sump



07a00131

Use the above charts to calculate the oil change interval depending on the lubrication oil sump.